

**WEB DEVELOPMENT**  
**PROJECT REPORT**  
**Library Management System (SMARTLIB)**

## **TABLE OF CONTENTS**

1. Introduction
2. Objectives of the System
3. Software and Tools Used
4. User Roles in SmartLib
5. Functional Requirements
  - 5.1 Welcome Page
  - 5.2 Login System
  - 5.3 Session and Cookie Management
  - 5.4 Student Module
    - 5.4.1 Student Dashboard
    - 5.4.2 Browse Books
    - 5.4.3 My Books
    - 5.4.4 Request Book
    - 5.4.5 Return Book
    - 5.4.6 Profile Management
  - 5.5 Faculty Module
    - 5.5.1 Faculty Dashboard
    - 5.5.2 Issue Books
    - 5.5.3 Return Books
    - 5.5.4 Recommended Books
    - 5.5.5 Priority Book Requests
    - 5.5.6 Reports
    - 5.5.7 Faculty Profile Management
  - 5.6 Librarian Module
    - 5.6.1 Librarian Dashboard
    - 5.6.2 Book Management
    - 5.6.3 Member Management
    - 5.6.4 Issue & Return Books
    - 5.6.5 Reservations (Book Requests)
    - 5.6.6 Reports & Analytics
    - 5.6.7 Account Settings
6. Non-Functional Requirements
  - 6.1 User-Friendly Interface
  - 6.2 Secure Login and Session Management
  - 6.3 Role-Based Access Control
  - 6.4 Fast Response Time
  - 6.5 Data Accuracy and Consistency
7. Conclusion

## **1. Introduction**

SmartLib is a web-based Library Management System designed to automate and simplify library operations. The system allows students, faculty members, and librarians to interact with the library through a single online platform. It manages book records, issuing and returning of books, user requests, fines, and reports efficiently.

## **2. Objectives of the System**

The main objectives of SmartLib are:

- To automate manual library processes
- To maintain accurate records of books and users
- To provide role-based access (Student, Faculty, Librarian)
- To manage book issuing, returning, and overdue tracking
- To reduce paperwork and human errors

## **3. Software and Tools Used**

- Visual Studio Code (Code Editor)
- XAMPP (Apache Server & MySQL)
- PHP (Server-side scripting)
- MySQL / phpMyAdmin (Database)
- HTML, CSS (User Interface Design)

## **4. User Roles in SmartLib**

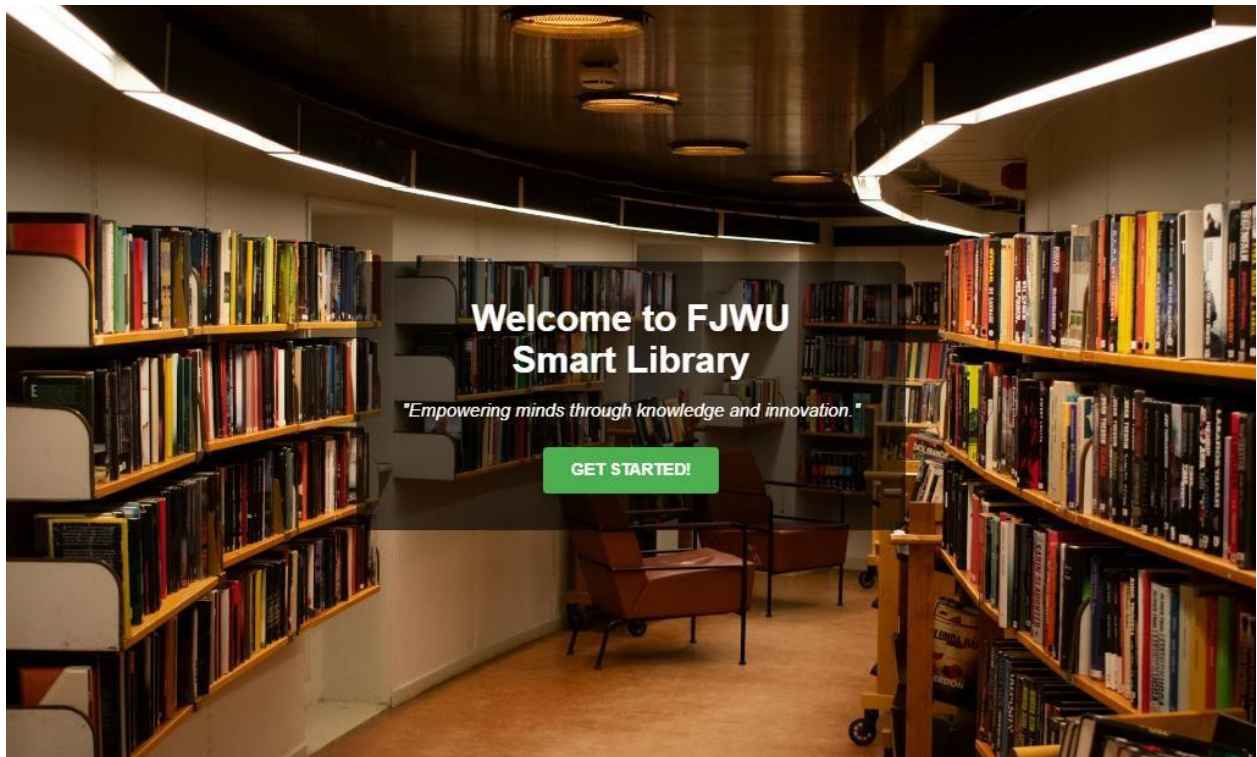
SmartLib supports three main user roles:

- Student
- Faculty
- Librarian

## 5. Functional Requirements

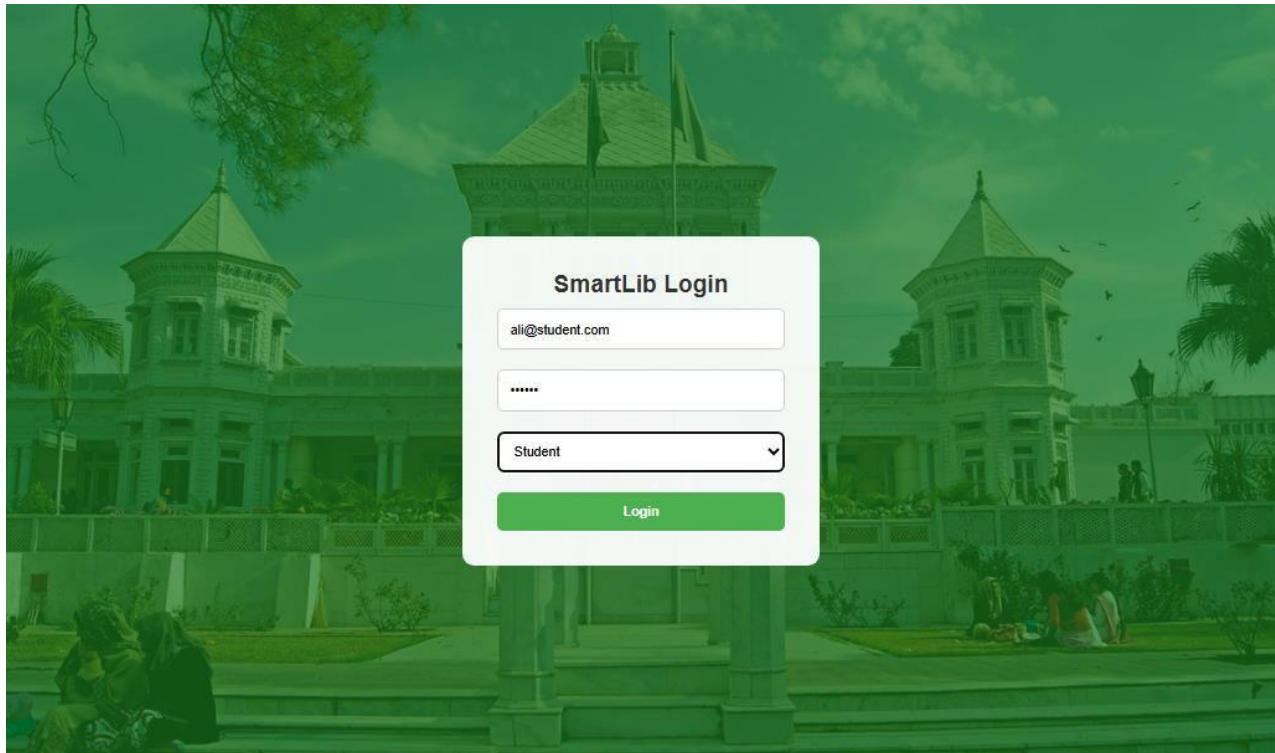
### 5.1 Welcome Page

The welcome page is the first interface of the system. It displays a brief introduction to SmartLib and provides a **Get Started** button that redirects users to the login page.



### 5.2 Login System

Users log in using their email, password, and role selection. After successful authentication, the system redirects them to their respective dashboards.



## 5.3 Session and Cookie Management

The SmartLib system uses **PHP Sessions** and **Cookies** to manage user authentication and improve user experience.

Sessions are used to store secure information such as `member_id` and user role after login. This ensures that only authorized users can access their respective dashboards. Session data is stored on the server and is destroyed on logout or browser close.

Cookies are used for non-sensitive purposes such as remembering the user's email address and storing last visit information. Cookies remain stored on the user's browser for a limited time to improve usability without compromising security.

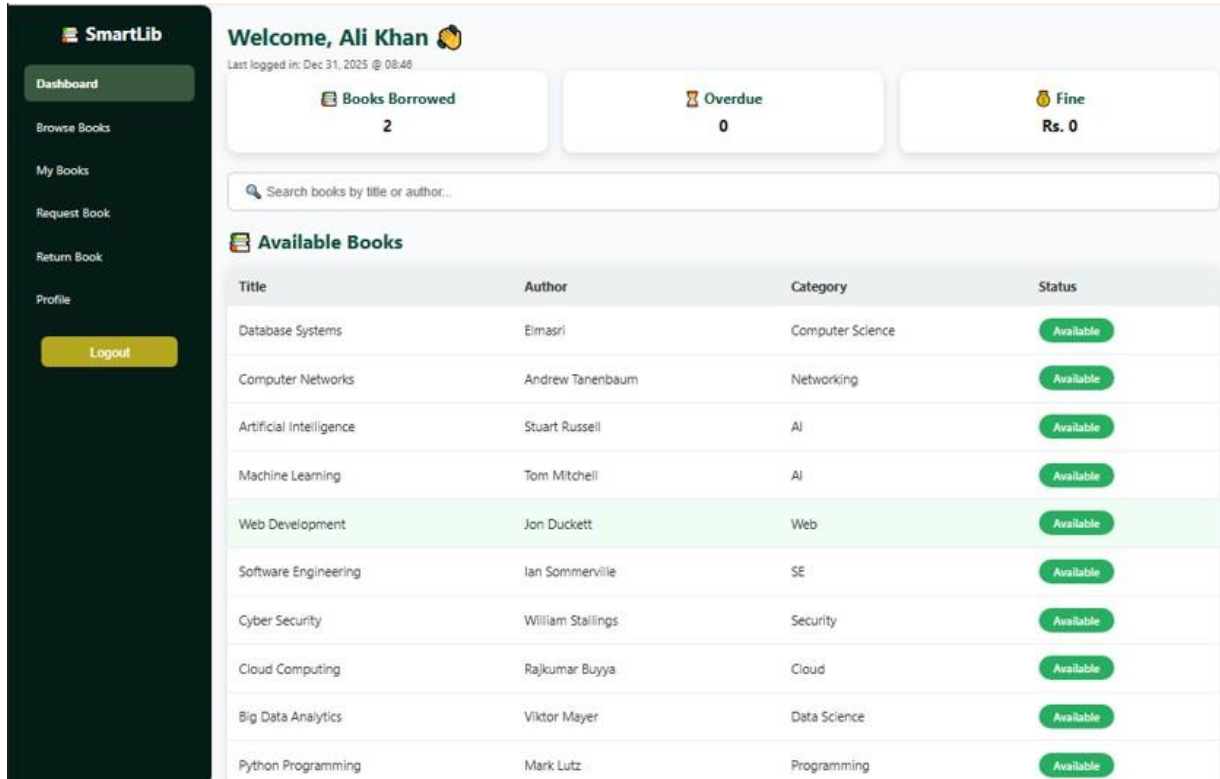
## 5.4 Student Module

### 5.4.1 Student Dashboard

The Student Dashboard is the main screen after login. It provides an overview of the student's library activity.

## Features shown on dashboard:

- Welcome message with last login time
- Total Books Borrowed
- Overdue Books Count
- Total Fine Amount



The screenshot shows the SmartLib dashboard for a user named Ali Khan. The dashboard includes a sidebar with navigation links: Dashboard, Browse Books, My Books, Request Book, Return Book, and Profile. The main content area displays a welcome message, last login time (Dec 31, 2025 @ 08:46), and three summary cards: Books Borrowed (2), Overdue (0), and Fine (Rs. 0). Below these is a search bar and a table of available books.

| Title                   | Author            | Category         | Status    |
|-------------------------|-------------------|------------------|-----------|
| Database Systems        | Eimasri           | Computer Science | Available |
| Computer Networks       | Andrew Tanenbaum  | Networking       | Available |
| Artificial Intelligence | Stuart Russell    | AI               | Available |
| Machine Learning        | Tom Mitchell      | AI               | Available |
| Web Development         | Jon Duckett       | Web              | Available |
| Software Engineering    | Ian Sommerville   | SE               | Available |
| Cyber Security          | William Stallings | Security         | Available |
| Cloud Computing         | Rajkumar Buyya    | Cloud            | Available |
| Big Data Analytics      | Viktor Mayer      | Data Science     | Available |
| Python Programming      | Mark Lutz         | Programming      | Available |

This dashboard helps students quickly understand their current library status.

### 5.4.2 Browse Books

The Browse Books page allows students to:

- Search books by title, author, or ISBN
- Filter books by category and availability
- View book details such as title, author, category, ISBN, copies, and status
- Submit a request for available books

- Dashboard
- Browse Books**
- My Books
- Request Book
- Return Book
- Profile
- Logout

## Browse Books

Find books by searching or filtering below

All Categories

All Status

### Library Catalog

| Title                   | Author            | Category         | ISBN    | Copies | Status    | Action      |
|-------------------------|-------------------|------------------|---------|--------|-----------|-------------|
| Artificial Intelligence | Stuart Russell    | AI               | ISBN004 | 0      | AVAILABLE | Unavailable |
| Big Data Analytics      | Viktor Mayer      | Data Science     | ISBN011 | 1      | AVAILABLE | Request     |
| Cloud Computing         | Rajkumar Buyya    | Cloud            | ISBN009 | 3      | AVAILABLE | Request     |
| Compiler Design         | Aho Ullman        | Computer Science | ISBN015 | 1      | REQUESTED | Request     |
| Computer Networks       | Andrew Tanenbaum  | Networking       | ISBN003 | 3      | AVAILABLE | Request     |
| Cyber Security          | William Stallings | Security         | ISBN008 | 3      | AVAILABLE | Request     |
| Data Mining             | Han & Kamber      | Data Science     | ISBN010 | 2      | REQUESTED | Request     |
| Database Systems        | Elmasri           | Computer Science | ISBN001 | 5      | AVAILABLE | Request     |
| Discrete Mathematics    | Kenneth Rosen     | Mathematics      | ISBN014 | 3      | AVAILABLE | Request     |
| Java Complete Reference | Herbert Schildt   | Programming      | ISBN013 | 4      | REQUESTED | Request     |

### 5.4.3 My Books

This section displays the complete book history of the student, including:

- Issued books
- Returned books
- Due dates
- Status (Issued / Returned / Overdue)
- Fine (if applicable)

- Dashboard
- Browse Books
- My Books**
- Request Book
- Return Book
- Profile
- Logout

## My Books

Track your borrowed books and request status

### Your Book History

| Title                   | Author          | Issue Date | Due Date   | Status           | Fine |
|-------------------------|-----------------|------------|------------|------------------|------|
| Machine Learning        | Tom Mitchell    | 2025-12-31 | 2026-01-14 | RETURNED         | -    |
| Python Programming      | Mark Lutz       | 2025-12-31 | 2026-01-14 | RETURNED         | -    |
| Java Complete Reference | Herbert Schildt | 2025-12-31 | 2026-01-14 | ISSUED / ON TIME | -    |
| Operating Systems       | Silberschatz    | 2025-12-31 | 2026-01-14 | ISSUED / ON TIME | -    |
| Artificial Intelligence | Stuart Russell  | 2025-02-01 | 2025-02-20 | RETURNED         | -    |
| Database Systems        | Elmasri         | 2025-01-01 | 2025-01-15 | RETURNED         | -    |

### 5.4.4 Request Book

Students can submit a book request by providing:

- Book title
- Author name
- Category

After submission, the request appears in **My Previous Requests** with status **Pending** until approved by the librarian. If both a student and a faculty request the same book, the faculty request is given priority.

| Book Title       | Author  | Request Date | Status  |
|------------------|---------|--------------|---------|
| Database Systems | Elmasri | 2025-12-31   | Pending |

### 5.4.5 Return Book

Students can return issued books from this section.

#### Functions:

- Select issued book from dropdown
- Submit return request
- Automatically update return date and status
- View return history

SmartLib

Dashboard

Browse Books

My Books

Request Book

Return Book

Profile

Logout

Return a Book

Select the book you want to return ☐

✔ Book returned successfully!

Select Book to Return

-- Choose Your Issued Book --

Return Book

Return History

| Title                   | Return Date | Status   |
|-------------------------|-------------|----------|
| Machine Learning        | 2025-12-31  | RETURNED |
| Python Programming      | 2025-12-31  | RETURNED |
| Java Complete Reference | 2025-12-31  | RETURNED |
| Artificial Intelligence | 2025-12-30  | RETURNED |

### 5.4.6 Profile Management

Students can:

- Update personal information (name, email, phone, address)
- Change password
- Delete account permanently

SmartLib

Dashboard

Browse Books

My Books

Request Book

Return Book

Profile

Logout

My Profile

Manage your account information

Personal Information

Full Name

Ali Khan

Email Address

ali@student.com

Phone Number

03001234567

Address

Lahore

Password

ali123

Save Changes

Account Settings

Want to delete your account permanently?

Delete Account

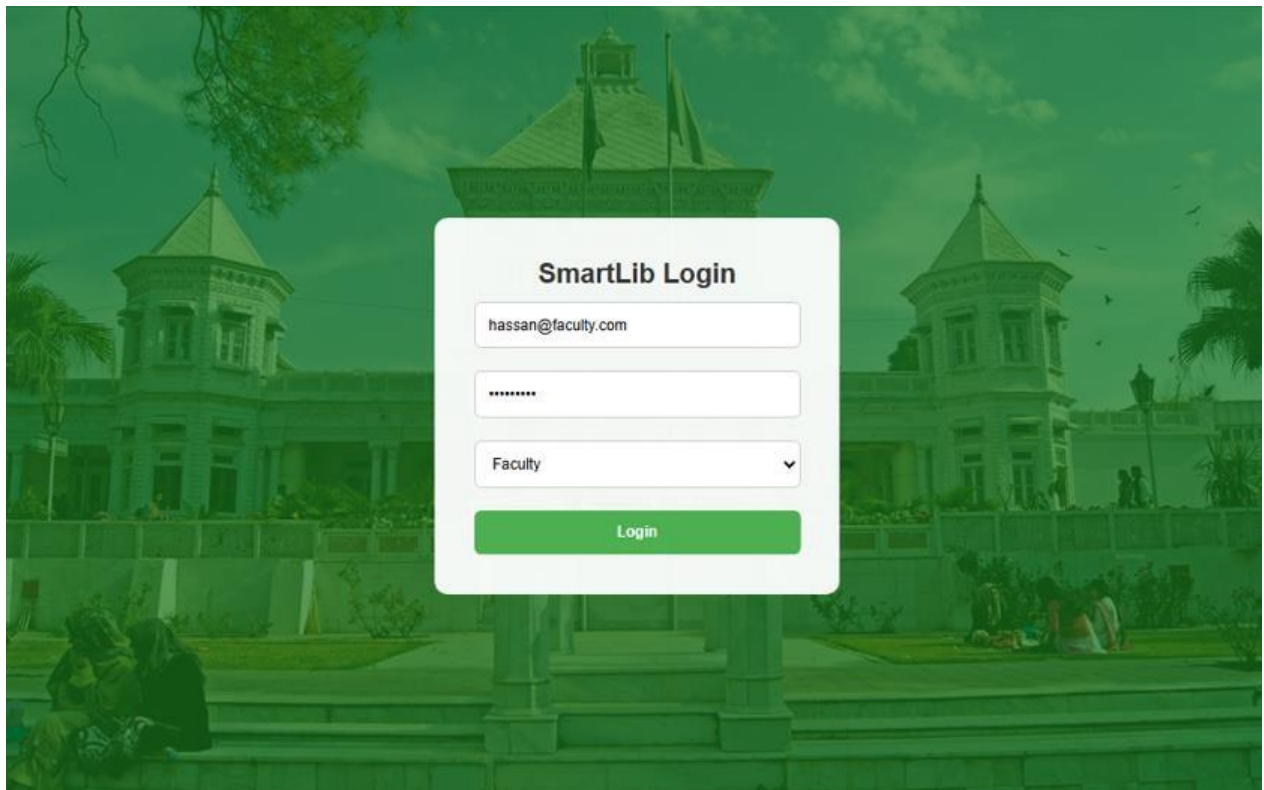
## 5.5 Faculty Module

Faculty members have similar features to students but with additional privileges:

- Priority access in book requests
- Extended issue duration (up to 30 days)
- No overdue fine policy
- Book recommendation feature

### 5.5.1 Faculty Dashboard

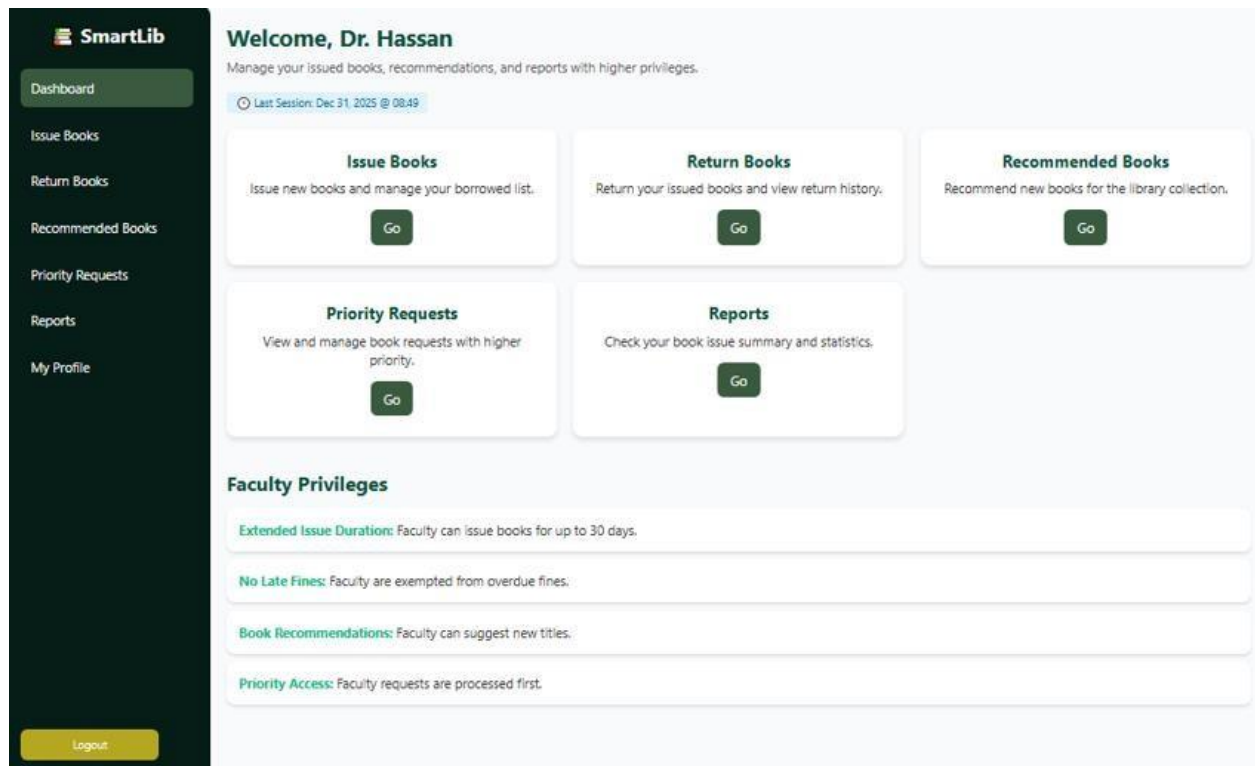
After successful login, faculty members are redirected to the **Faculty Dashboard**.



#### Dashboard Features:

- Welcome message with last login time
- Quick access buttons for:
  - Issue Books
  - Return Books
  - Recommended Books
  - Priority Requests

- Reports
- Display of faculty privileges



The dashboard acts as a central control panel for all faculty-related operations.

### 5.5.2 Issue Books

Faculty members can issue books directly from the **Issue Books** section.

#### Functional Details:

- Faculty can issue books for **up to 30 days**
- Book selection through dropdown list
- Issue duration selectable (default: 30 days)
- Issued book appears in issued history with status **Issued**

- Dashboard
- Issue Books
- Return Books
- Recommended Books
- Priority Requests
- Reports
- My Profile

### Issue Books

Faculty can issue books for up to 30 days.

☒ Book requested successfully!

Book ID

Select Book

Issue Duration (Days)

30 Days

Issue Book

| Title                   | Author            | Issue Date | Due Date   | Return Date | Status    |
|-------------------------|-------------------|------------|------------|-------------|-----------|
| Cyber Security          | William Stallings | 2025-12-31 | 2026-01-30 | 2025-12-31  | Returned  |
| Big Data Analytics      | Viktor Mayer      | 2025-12-31 | 2026-01-30 | 2025-12-31  | Returned  |
| Web Development         | Jon Duckett       | 2025-12-31 | 2026-01-30 | —           | Issued    |
| Discrete Mathematics    | Kenneth Rosen     | 2025-12-31 | 2026-01-30 | —           | Requested |
| Artificial Intelligence | Stuart Russell    | 2025-12-30 | 2026-01-29 | 2025-12-30  | Returned  |
| Artificial Intelligence | Stuart Russell    | 2025-12-30 | 2026-01-29 | 2025-12-30  | Returned  |
| Web Development         | Jon Duckett       | 2025-02-05 | 2025-02-25 | 2025-12-30  | Returned  |

This feature allows faculty members extended access to books compared to students.

### 5.5.3 Return Books

Faculty members can return issued books at any time using the **Return Books** section.

#### Functions:

- Select issued book from dropdown
- Submit return request
- Automatically update return date
- Book status changes to **Returned**
- Return history is updated instantly

SmartLib

Dashboard

Issue Books

Return Books

Recommended Books

Priority Requests

Reports

My Profile

Logout

Return Books

Faculty members can return issued books anytime

Book returned successfully!

Select Issued Book

-- Select Issued Book --

Return Book

| Book Title              | Return Date |
|-------------------------|-------------|
| Web Development         | 2025-12-31  |
| Cyber Security          | 2025-12-31  |
| Big Data Analytics      | 2025-12-31  |
| Web Development         | 2025-12-31  |
| Artificial Intelligence | 2025-12-30  |
| Artificial Intelligence | 2025-12-30  |

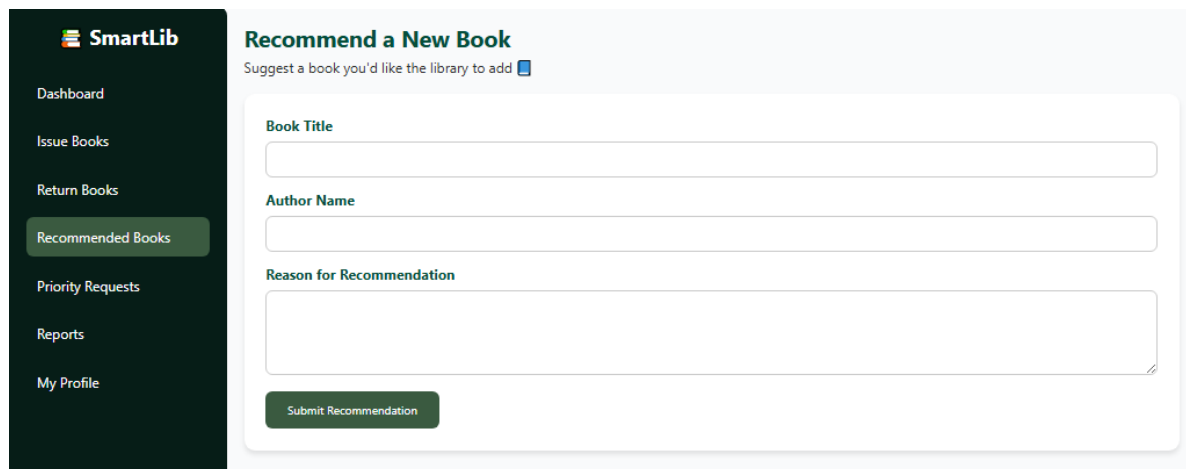
Faculty members are **exempted from late fines**, even if the book is returned after the due date.

#### 5.5.4 Recommended Books

Faculty members can suggest new books to be added to the library collection.

#### Recommendation Form Includes:

- Book title
- Author name
- Reason for recommendation



The image shows a web interface for 'SmartLib' with a sidebar menu on the left containing: Dashboard, Issue Books, Return Books, Recommended Books (highlighted), Priority Requests, Reports, and My Profile. The main content area is titled 'Recommend a New Book' with the subtitle 'Suggest a book you'd like the library to add'. It contains three input fields: 'Book Title', 'Author Name', and 'Reason for Recommendation' (a larger text area). A green 'Submit Recommendation' button is at the bottom.

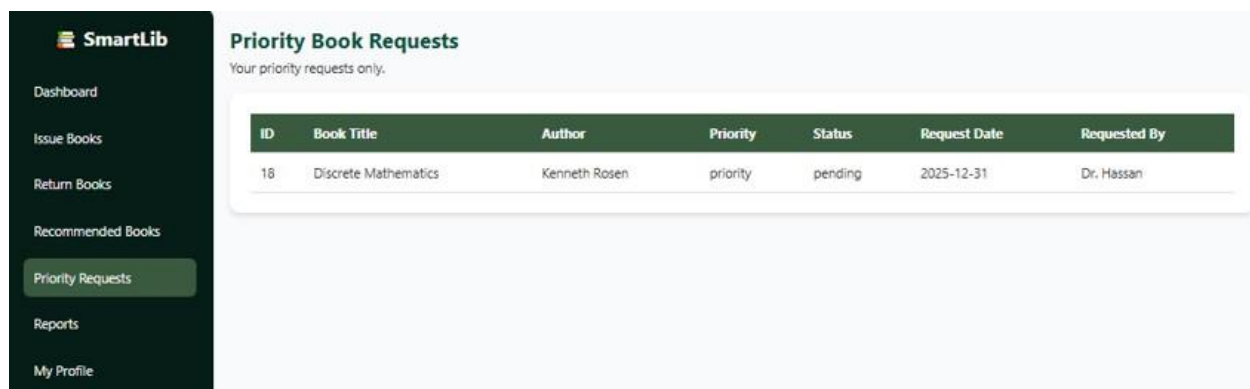
This feature helps improve the quality of the library collection.

### 5.5.5 Priority Book Requests

Faculty members have access to a **Priority Requests** section.

#### Priority Logic:

- Faculty book requests are marked as **Priority**
- If both a student and a faculty request the same book:
  - Faculty request is processed first



The image shows a web interface for 'SmartLib' with a sidebar menu on the left containing: Dashboard, Issue Books, Return Books, Recommended Books, Priority Requests (highlighted), Reports, and My Profile. The main content area is titled 'Priority Book Requests' with the subtitle 'Your priority requests only.' It displays a table with the following data:

| ID | Book Title           | Author        | Priority | Status  | Request Date | Requested By |
|----|----------------------|---------------|----------|---------|--------------|--------------|
| 18 | Discrete Mathematics | Kenneth Rosen | priority | pending | 2025-12-31   | Dr. Hassan   |

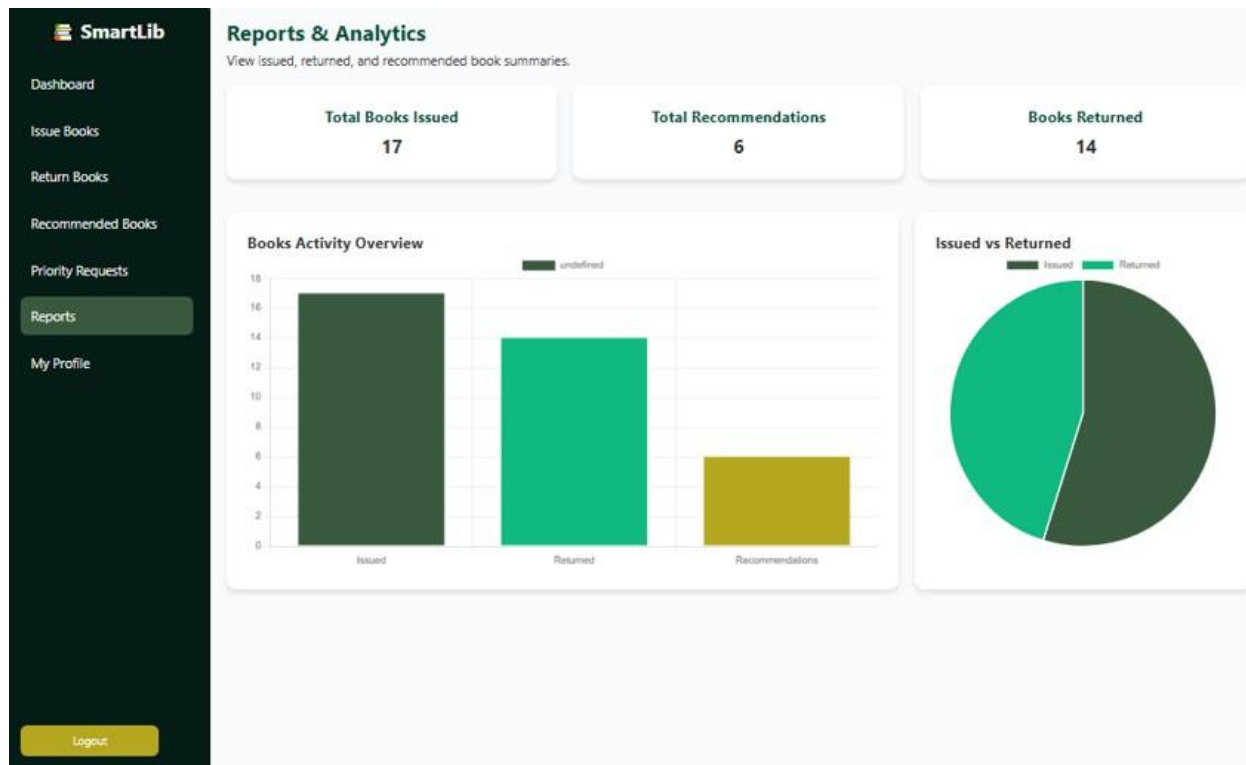
This ensures faculty members receive faster access to required books.

### 5.5.6 Reports

Faculty members can view analytical reports related to their library activity.

#### Reports Include:

- Total books issued
- Total books returned
- Total recommendations submitted
- Graphical representation (bar charts & pie charts)



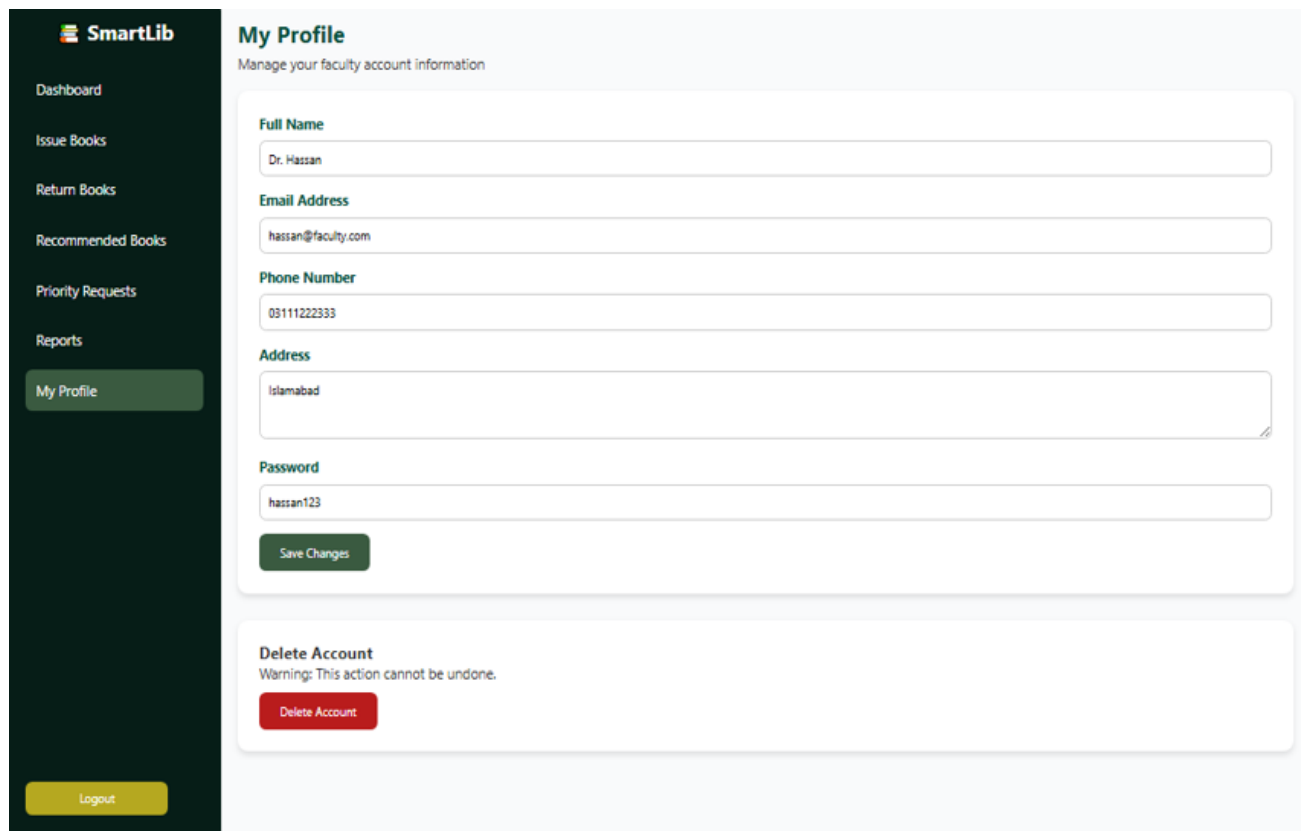
Reports help faculty track their usage and contribution to the library.

### 5.5.7 Faculty Profile Management

Faculty members can manage their personal profile information.

#### Profile Features:

- Update name, email, phone number, and address
- Change password
- Delete account permanently (irreversible action)



The image shows a web application interface for 'SmartLib'. On the left is a dark green sidebar with a list of navigation items: 'Dashboard', 'Issue Books', 'Return Books', 'Recommended Books', 'Priority Requests', 'Reports', 'My Profile' (which is highlighted with a green background), and a 'Logout' button at the bottom. The main content area is titled 'My Profile' with the subtitle 'Manage your faculty account information'. It contains several form fields: 'Full Name' (with 'Dr. Hassan'), 'Email Address' (with 'hassan@faculty.com'), 'Phone Number' (with '03111222333'), 'Address' (with 'Islamabad'), and 'Password' (with 'hassan123'). A green 'Save Changes' button is located below the password field. At the bottom of the main content area, there is a 'Delete Account' section with a warning message 'Warning: This action cannot be undone.' and a red 'Delete Account' button.

**SmartLib**

Dashboard

Issue Books

Return Books

Recommended Books

Priority Requests

Reports

**My Profile**

Logout

### My Profile

Manage your faculty account information

**Full Name**

Dr. Hassan

**Email Address**

hassan@faculty.com

**Phone Number**

03111222333

**Address**

Islamabad

**Password**

hassan123

Save Changes

**Delete Account**

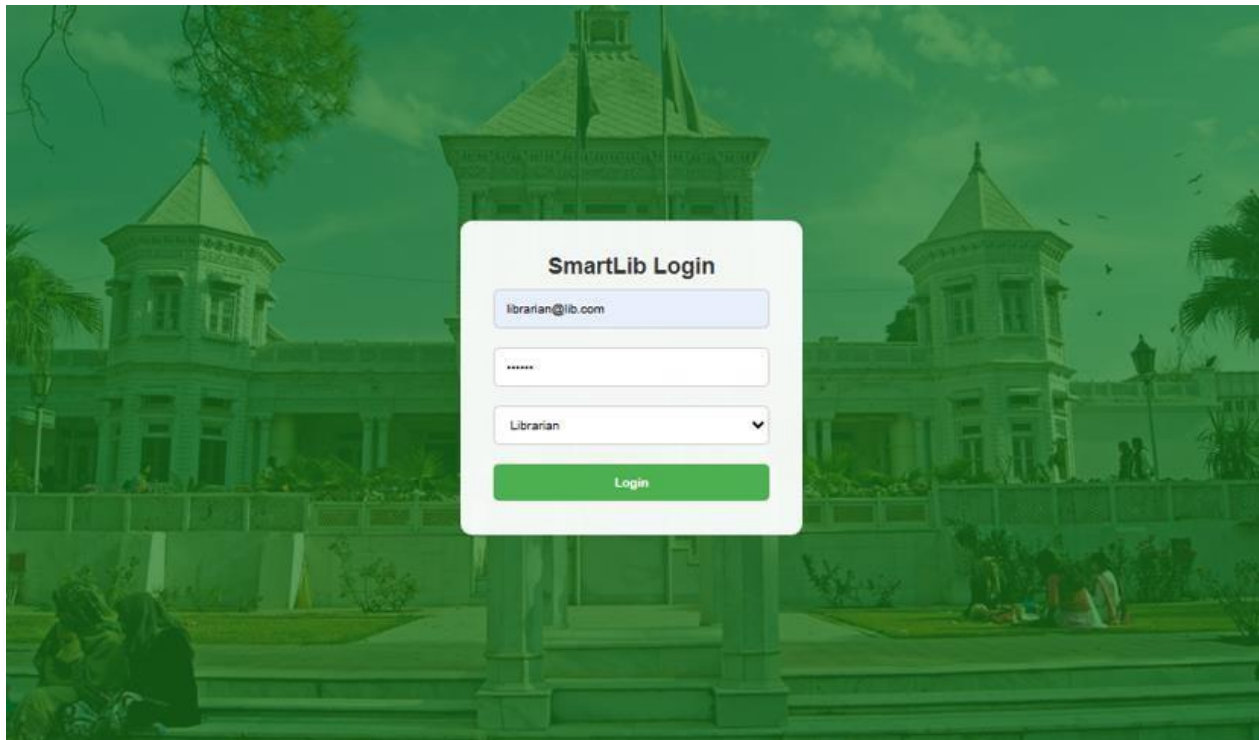
Warning: This action cannot be undone.

Delete Account

## 5.6 Librarian Module

The librarian dashboard provides complete control over the system

The librarian accesses the system through the SmartLib Login page by selecting the **Librarian** role and entering valid credentials.

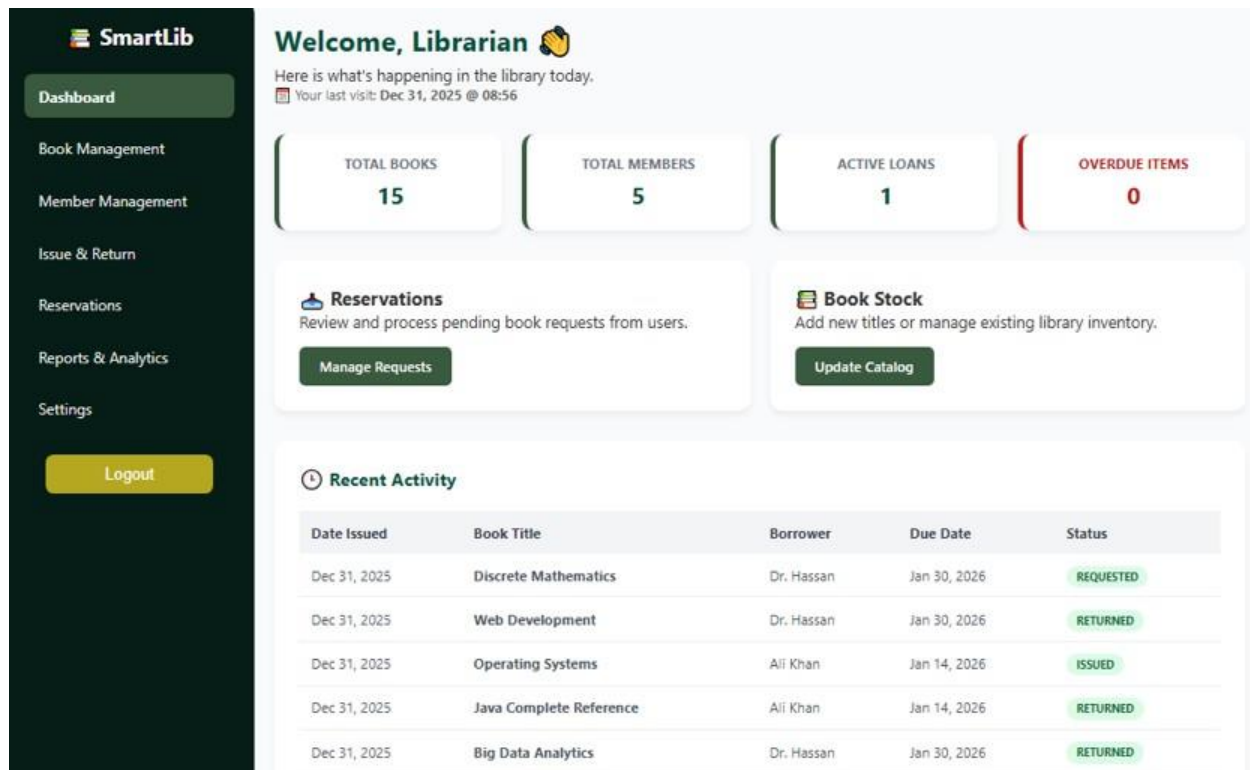


### 5.6.1 Librarian Dashboard

The Librarian Dashboard provides a real-time overview of library activity.

#### Dashboard Information Cards:

- Welcome message with last login time
- Total Books
- Total Members
- Active Loans
- Overdue Items
- Quick access buttons for Reservations and Book Stock
- Recent Activity table showing issued, returned, and requested books



This dashboard helps the librarian monitor library status at a glance.

### 5.6.2 Book Management

The Book Management module allows librarians to manage the library inventory.

Functions:

- Add new books (Title, Author, Category, ISBN, Copies)
- Modify existing book details
- Delete books from the catalog
- Search books by Title, Author, or ISBN
- View availability status (Available / Requested)

- Dashboard
- Book Management**
- Member Management
- Issue & Return
- Reservations
- Reports & Analytics
- Settings
- Logout

## Book Management

Add, Modify, or Delete books from the system.

+ Add Book
✎ Modify Book
🗑 Delete Book

Search

| ID | Title                   | Author            | Category         | ISBN    | Copies | Status    |
|----|-------------------------|-------------------|------------------|---------|--------|-----------|
| 1  | Database Systems        | Elmasri           | Computer Science | ISBN001 | 5      | Requested |
| 2  | Operating Systems       | Silberschatz      | Computer Science | ISBN002 | 4      | Requested |
| 3  | Computer Networks       | Andrew Tanenbaum  | Networking       | ISBN003 | 3      | Available |
| 4  | Artificial Intelligence | Stuart Russell    | AI               | ISBN004 | 0      | Available |
| 5  | Machine Learning        | Tom Mitchell      | AI               | ISBN005 | 2      | Available |
| 6  | Web Development         | Jon Duckett       | Web              | ISBN006 | 5      | Available |
| 7  | Software Engineering    | Ian Sommerville   | SE               | ISBN007 | 3      | Available |
| 8  | Cyber Security          | William Stallings | Security         | ISBN008 | 3      | Available |
| 9  | Cloud Computing         | Rajkumar Buyya    | Cloud            | ISBN009 | 3      | Available |
| 10 | Data Mining             | Han & Kamber      | Data Science     | ISBN010 | 2      | Requested |
| 11 | Big Data Analytics      | Viktor Mayer      | Data Science     | ISBN011 | 1      | Available |

### 5.6.3 Member Management

The Member Management module is used to manage users of the system.

Functions:

- Add new members
- Modify member details (Name, Email, Role)
- Delete members from the system
- View complete member registry
- Search members by Name, Email, or ID

SmartLib

Dashboard

Book Management

Member Management

Issue & Return

Reservations

Reports & Analytics

Settings

Logout

Member Management

Manage library access for students, faculty, and staff.

+ Add New Member

Full Name

Email

Phone

Address

Student

Add Member

✎ Modify Member

ID

New Name

New Email

Change Role (Optional)

Update

🗑 Delete Member

Enter Member ID

Delete

📋 Member Registry

Search by Name, Email, or ID...

Search

| ID | Name       | Email              | Role      | Date Joined  |
|----|------------|--------------------|-----------|--------------|
| #5 | Librarian  | librarian@lib.com  | LIBRARIAN | Dec 30, 2025 |
| #4 | Dr. Alina  | alina@faculty.com  | FACULTY   | Dec 30, 2025 |
| #3 | Dr. Hassan | hassan@faculty.com | FACULTY   | Dec 30, 2025 |
| #2 | Sara Ahmed | sara@student.com   | STUDENT   | Dec 30, 2025 |
| #1 | Ali Khan   | ali@student.com    | STUDENT   | Dec 30, 2025 |

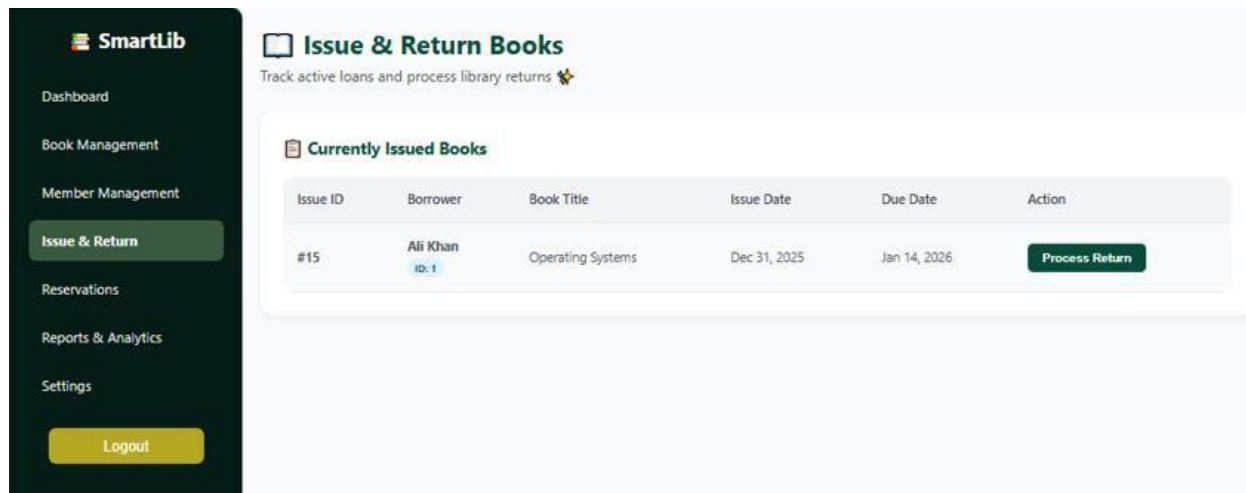
This module helps librarians control access and maintain user records.

## 5.6.4 Issue & Return Books

This module handles book issuing and returning operations.

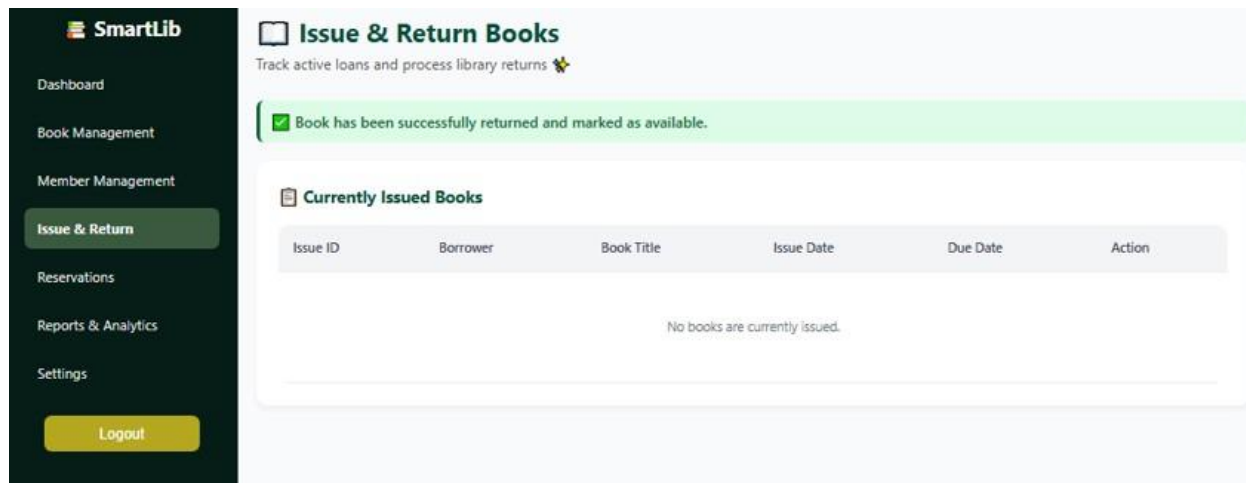
Issue Books:

- View currently issued books
- Track borrower details
- Monitor issue date and due date



Return Books:

- Process book returns
- Automatically update book status to Available
- Update return history
- Display success confirmation messages



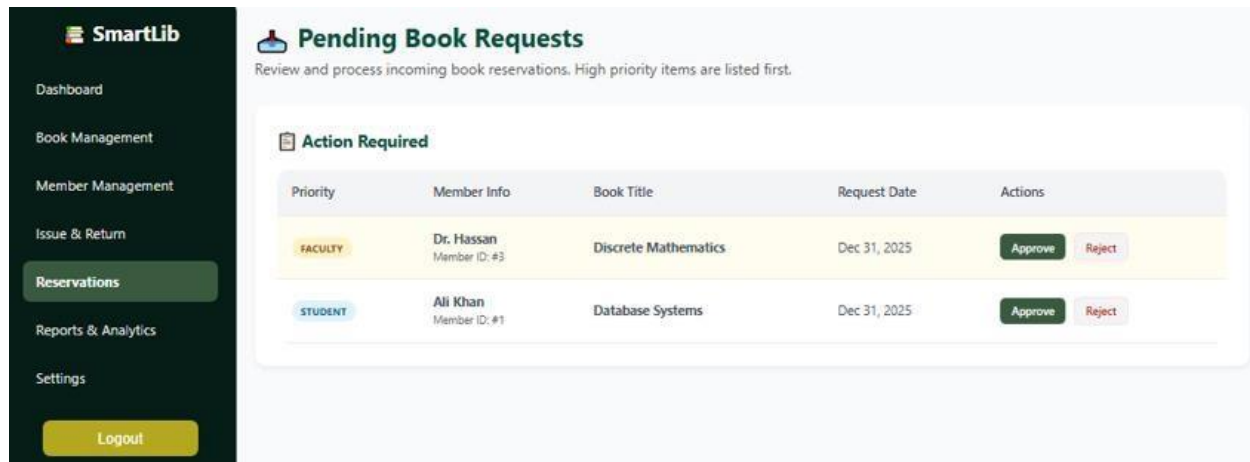
### 5.6.5 Reservations (Book Requests)

The Reservations module manages pending book requests.

Features:

- View all pending book requests
- Faculty requests are marked as **High Priority**
- Approve or reject requests

- Automatically assign due dates upon approval



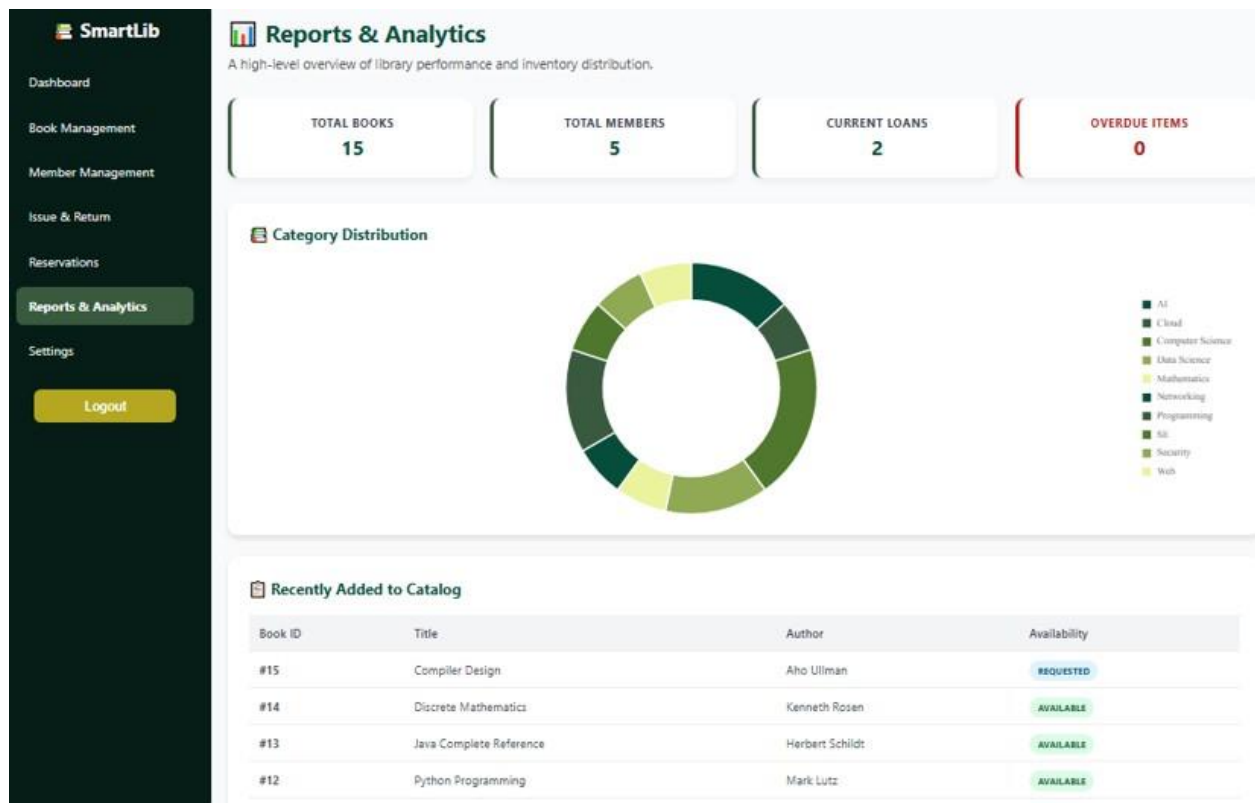
**Priority Rule:** If both a Student and a Faculty member request the same book, the Faculty request is processed first.

### 5.6.6 Reports & Analytics

The Reports & Analytics module provides insights into library performance.

Reports Include:

- Total Books
- Total Members
- Current Loans
- Overdue Items
- Category-wise book distribution (graphical view)
- Recently added books



This module helps librarians analyze usage patterns and inventory distribution.

### 5.6.7 Account Settings

The Settings module allows librarians to manage their account.

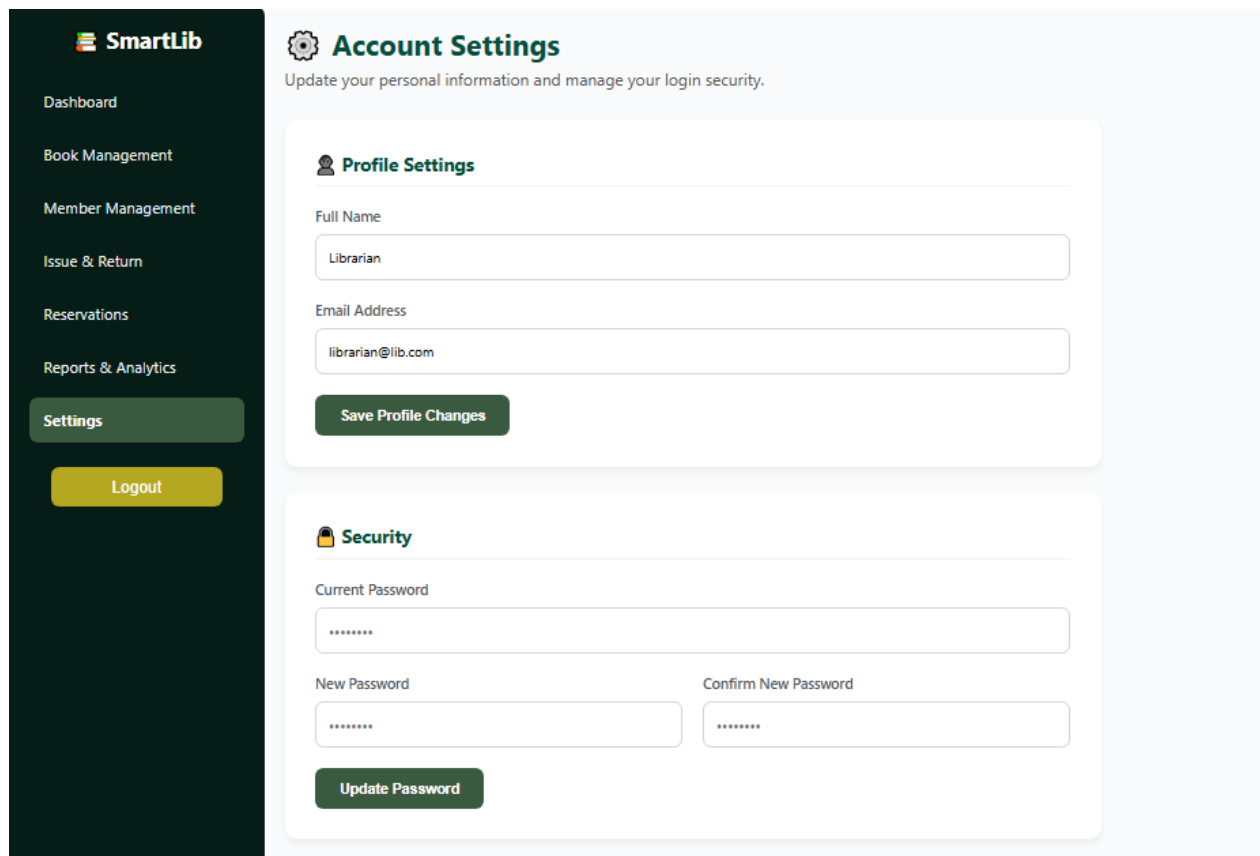
Profile Settings:

- Update full name
- Update email address

Security Settings:

- Change password
- Confirm password update securely

This ensures account security and updated personal information.

The image shows a web application interface for 'SmartLib'. On the left is a dark green sidebar with a 'SmartLib' logo at the top. Below the logo are several menu items: 'Dashboard', 'Book Management', 'Member Management', 'Issue & Return', 'Reservations', 'Reports & Analytics', 'Settings' (highlighted with a green background), and 'Logout' (highlighted with a yellow background). The main content area is white and titled 'Account Settings' with a gear icon. Below the title is a subtitle: 'Update your personal information and manage your login security.' There are two main sections: 'Profile Settings' and 'Security'. The 'Profile Settings' section has a header with a person icon, followed by a 'Full Name' field containing 'Librarian', an 'Email Address' field containing 'librarian@lib.com', and a green 'Save Profile Changes' button. The 'Security' section has a header with a lock icon, followed by a 'Current Password' field with masked characters, two 'New Password' fields (one with masked characters and one empty), and a 'Confirm New Password' field with masked characters. A green 'Update Password' button is at the bottom of the security section.

## 6. Non-Functional Requirements

Non-functional requirements describe how the system performs its functions rather than what functions it performs. These requirements ensure that the SmartLib system is reliable, secure, and easy to use.

### 6.1 User-Friendly Interface

The SmartLib system provides a clean, simple, and well-organized user interface. Dashboards are designed according to user roles (Student, Faculty, Librarian), allowing users to easily understand and navigate the system. Clear buttons, readable fonts, and meaningful labels help users perform tasks without technical knowledge or training.

### 6.2 Secure Login and Session Management

Security is a key requirement of the SmartLib system. User authentication is handled through a secure login mechanism using email, password, and role selection. PHP sessions are used to maintain user login states and prevent

unauthorized access. Users cannot access restricted pages without proper authentication, and session termination occurs on logout to ensure safety.

In addition to sessions, cookies are used for non-sensitive data such as last login information and remembered email address. This combination of sessions and cookies ensures strong security while providing a better user experience.

### **6.3 Role-Based Access Control**

SmartLib implements strict role-based access control to protect system data. Each user role has specific permissions:

- Students can only access student-related features.
  - Faculty members have additional privileges such as priority requests and extended issue duration.
  - Librarians have full administrative access.
- This separation of roles ensures data integrity and prevents misuse of system resources.

### **6.4 Fast Response Time**

The system is designed to respond quickly to user actions such as searching books, submitting requests, and loading dashboards. Efficient database queries and optimized backend logic help reduce delays, ensuring a smooth user experience even with multiple users accessing the system simultaneously.

### **6.5 Data Accuracy and Consistency**

SmartLib ensures accurate and consistent data by using a centralized MySQL database. All records related to books, users, issues, returns, and fines are stored and updated in real time. Validation checks are applied to avoid incorrect data entry, ensuring reliable reports and correct system behavior.

## **7. Conclusion**

SmartLib is a complete and efficient web-based Library Management System designed to automate and simplify library operations. The system successfully replaces manual processes with a digital platform that improves accuracy, saves time, and reduces paperwork.

By providing role-based access for students, faculty, and librarians, SmartLib ensures secure and organized management of library activities. Features such as book requests, priority handling for faculty, issue and return management, fine calculation, and reporting make the system practical and effective for real-world use.

Overall, SmartLib enhances the productivity of librarians, improves accessibility for users, and offers a reliable solution for modern library management.